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Java SE 27 & JDK 27

Class Channels

java.lang.Object
    java.nio.channels.Channels

```
public final class Channels
extends Object
```

Utility methods for channels and streams.

This class defines static methods that support the interoperation of the stream classes of the `java.io` package with the channel classes of this package.

Since:

1.4

Method Summary

All Methods	Static Methods	

Modifier and Type	Method	Description
static ReadableByteChannel	**newChannel** (InputStream in)	Constructs a channel that reads bytes from the given stream.
static WritableByteChannel	**newChannel** (OutputStream out)	Constructs a channel that writes bytes to the given stream.
static InputStream	**newInputStream** (AsynchronousByteChannel	Constructs a stream that reads bytes from the given channel.
static InputStream	**newInputStream** (ReadableByteChannel ch)	Constructs a stream that reads bytes from the given channel.
static OutputStream	**newOutputStream** (AsynchronousByteChannel	Constructs a stream that writes bytes to the given channel.
static OutputStream	**newOutputStream** (WritableByteChannel ch)	Constructs a stream that writes bytes to the given channel.
static Reader	**newReader** (ReadableByteChannel ch, String csName)	Constructs a reader that decodes bytes from the

		given channel according to the named charset.
static Reader	**newReader**(ReadableByteChannel ch, Charset charset)	Constructs a reader that decodes bytes from the given channel according to the given charset.
static Reader	**newReader**(ReadableByteChannel ch, CharsetDecoder dec, int minBufferCap)	Constructs a reader that decodes bytes from the given channel using the given decoder.
static Writer	**newWriter**(WritableByteChannel ch, String csName)	Constructs a writer that encodes characters according to the named charset and writes the resulting bytes to the given channel.
static Writer	**newWriter**(WritableByteChannel ch, Charset charset)	Constructs a writer that encodes characters according to the given charset and writes the resulting bytes to the given channel.
static Writer	**newWriter**(WritableByteChannel ch, CharsetEncoder enc, int minBufferCap)	Constructs a writer that encodes characters using the given encoder and writes the resulting bytes to the given channel.

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Method Details

newInputStream

```
public static InputStream newInputStream(ReadableByteChannel ch)
```

Constructs a stream that reads bytes from the given channel.

The `read` and `transferTo` methods of the resulting stream will throw an `IllegalBlockingModeException` if invoked while the underlying channel is in non-blocking mode. The `transferTo` method will also throw an `IllegalBlockingModeException` if invoked to transfer bytes to an output stream that writes to an underlying channel in non-blocking mode. The stream will not be buffered, and it will not support the `mark` or `reset` methods. The stream will be safe for access by multiple concurrent threads. Closing the stream will in turn cause the channel to be closed.

Parameters:

`ch` - The channel from which bytes will be read

Returns:

A new input stream

newOutputStream

```
public static OutputStream newOutputStream(WritableByteChannel ch)
```

Constructs a stream that writes bytes to the given channel.

The `write` methods of the resulting stream will throw an `IllegalBlockingModeException` if invoked while the underlying channel is in non-blocking mode. The stream will not be buffered. The stream will be safe for access by multiple concurrent threads. Closing the stream will in turn cause the channel to be closed.

Parameters:

`ch` - The channel to which bytes will be written

Returns:

A new output stream

newInputStream

```
public static InputStream newInputStream(AsynchronousByteChannel ch)
```

Constructs a stream that reads bytes from the given channel.

The stream will not be buffered, and it will not support the `mark` or `reset` methods. The stream will be safe for access by multiple concurrent threads. Closing the stream will in turn cause the channel to be closed.

Parameters:

`ch` - The channel from which bytes will be read

Returns:

A new input stream

Since:

1.7

newOutputStream

```
public static OutputStream newOutputStream
(AsynchronousByteChannel ch)
```

Constructs a stream that writes bytes to the given channel.

The stream will not be buffered. The stream will be safe for access by multiple concurrent threads. Closing the stream will in turn cause the channel to be closed.

Parameters:

`ch` - The channel to which bytes will be written

Returns:

A new output stream

Since:

1.7

newChannel

```
public static ReadableByteChannel newChannel(InputStream in)
```

Constructs a channel that reads bytes from the given stream.

The resulting channel will not be buffered; it will simply redirect its I/O operations to the given stream. Reading from the resulting channel will read from the input stream and thus block until input is available or end of file is reached. Closing the channel will in turn cause the stream to be closed.

Parameters:

`in` - The stream from which bytes are to be read

Returns:

A new readable byte channel

newChannel

```
public static WritableByteChannel newChannel(OutputStream out)
```

Constructs a channel that writes bytes to the given stream.

The resulting channel will not be buffered; it will simply redirect its I/O operations to the given stream. Closing the channel will in turn cause the stream to be closed.

Parameters:

`out` - The stream to which bytes are to be written

Returns:

A new writable byte channel

newReader

```
public static Reader newReader(ReadableByteChannel ch,
                               CharsetDecoder dec,
                               int minBufferCap)
```

Constructs a reader that decodes bytes from the given channel using the given decoder.

The resulting stream will contain an internal input buffer of at least `minBufferCap` bytes. The stream's `read` methods will, as needed, fill the buffer by reading bytes from the underlying channel; if the channel is in non-blocking mode when bytes are to be read then an `IllegalBlockingModeException` will be thrown. The resulting stream will not otherwise be buffered, and it will not support the `mark` or `reset` methods. Closing the stream will in turn cause the channel to be closed.

Parameters:

`ch` - The channel from which bytes will be read

`dec` - The charset decoder to be used

`minBufferCap` - The minimum capacity of the internal byte buffer, or `-1` if an implementation-dependent default capacity is to be used

Returns:

A new reader

newReader

```
public static Reader newReader(ReadableByteChannel ch,
                               String csName)
```

Constructs a reader that decodes bytes from the given channel according to the named charset.

An invocation of this method of the form

```
    Channels.newReader(ch, csname)
```

behaves in exactly the same way as the expression

```
    Channels.newReader(ch, Charset.forName(csName))
```

Parameters:

`ch` - The channel from which bytes will be read

`csName` - The name of the charset to be used

Returns:

A new reader

Throws:

`UnsupportedCharsetException` - If no support for the named charset is available in this instance of the Java virtual machine

newReader

```
public static Reader newReader(ReadableByteChannel ch,
                               Charset charset)
```

Constructs a reader that decodes bytes from the given channel according to the given charset.

An invocation of this method of the form

```
    Channels.newReader(ch, charset)
```

behaves in exactly the same way as the expression

```
    Channels.newReader(ch, charset.newDecoder(), -1)
```

The reader's default action for malformed-input and unmappable-character errors is to report them. When more control over the error handling is required, the constructor that takes a CharsetDecoder should be used.

Parameters:

`ch` - The channel from which bytes will be read

`charset` - The charset to be used

Returns:

A new reader

Since:

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newWriter

```
public static Writer newWriter(WritableByteChannel ch,
                               CharsetEncoder enc,
                               int minBufferCap)
```

Constructs a writer that encodes characters using the given encoder and writes the resulting bytes to the given channel.

The resulting stream will contain an internal output buffer of at least `minBufferCap` bytes. The stream's `write` methods will, as needed, flush the buffer by writing bytes to the underlying channel; if the channel is in non-blocking mode when bytes are to be written then an `IllegalBlockingModeException` will be thrown. The resulting stream will not otherwise be buffered. Closing the stream will in turn cause the channel to be closed.

Implementation Note:

The value of `minBufferCap` is ignored.

Parameters:

`ch` - The channel to which bytes will be written

`enc` - The charset encoder to be used

`minBufferCap` - The minimum capacity of the internal byte buffer, or `-1` if an implementation-dependent default capacity is to be used. The value of `minBufferCap` may be ignored

Returns:

A new writer

newWriter

```
public static Writer newWriter(WritableByteChannel ch,
                               String csName)
```

Constructs a writer that encodes characters according to the named charset and writes the resulting bytes to the given channel.

An invocation of this method of the form

```
    Channels.newWriter(ch, csname)
```

behaves in exactly the same way as the expression

```
    Channels.newWriter(ch, Charset.forName(csName))
```

Parameters:

`ch` - The channel to which bytes will be written

`csName` - The name of the charset to be used

Returns:

A new writer

Throws:

`UnsupportedCharsetException` - If no support for the named charset is available in this instance of the Java virtual machine

newWriter

```
public static Writer newWriter(WritableByteChannel ch,
                               Charset charset)
```

Constructs a writer that encodes characters according to the given charset and writes the resulting bytes to the given channel.

An invocation of this method of the form

```
    Channels.newWriter(ch, charset)
```

behaves in exactly the same way as the expression

```
    Channels.newWriter(ch, charset.newEncoder(), -1)
```

The writer's default action for malformed-input and unmappable-character errors is to report them. When more control over the error handling is required, the constructor that takes a CharsetEncoder should be used.

Parameters:

`ch` - The channel to which bytes will be written

`charset` - The charset to be used

Returns:

A new writer

Since:

10

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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